

Towards Energy Simulations for Proportionally Designed and Controlled Data Centers

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Abstract

Expressing site-specific load profiles and hardware specifications that reflect the information technology (IT) processes inside data centers have always been a challenge to do either benchmarks of building energy performance or predictive controls. As a substitute, data center building models usually assume naive hour-day-week schedules that don't align simulated operations with the weather profiles. This paper demonstrates a method towards using more representative load profiles in data center energy models by deriving the profiles based on global usage patterns of an internet platform. It is the first attempt to illustrate the coupling of an external signal with EnergyPlus as an indicator of IT load in a data center.

Introduction

This paper describes a data center (DC) building energy simulation framework. The framework is based on EnergyPlus as the physical model of the building operations and an externally implemented network traffic schedule as an indicator of the IT load. Namely, the target applications for these frameworks are the hyper-scale facilities that house global services such as social networks and search engines. For this class of facilities, the significance of network traffic to DC operations is exemplified by the sudden change in peoples internet usage behaviors due to the COVID-19 pandemic. Through the pandemic, DCs across the globe have seen a disruptive shift in user traffic patterns that were not known just weeks prior to the first stay-in-place orders. Given this timely example, it is imperative that DC operators be able to simulate their facilities to verify that their infrastructure can handle these types of shifts in traffic. However, presently there is a lack of publicly available information about how to build such simulations.

The contribution of this work is two-fold. First, it provides a demonstration for characterizing data centers and IT equipment in EnergyPlus. Second, by deriving the workload profile from network traffic, this work provides a perspective for reasoning about temporal profiles of the workloads that hyper-scale data centers must support.

DCs have 10 times the power density of conventional buildings. With this density, it is essential that during the design phase designers have access to representative building energy models to size equipment and to validate jurisdictional compliance. However, these building energy models are typically put on the shelf once the facility is turned over to the operational teams. With much of the life cycle costs of a data center attributed to the use-phase, having building energy models that represent the cyber-physical dynamics of their facility can be valuable to support operational decisions too.

In this work, wide area network traffic bandwidth is demonstrated as a proxy for IT loads in data center buildings. However, any technique that characterizes a fine grain coupling between the IT workloads and the building systems would be valuable for hyper-scale DCs the running online platforms such as social media sites, search engines, or public cloud services. These platforms are typically distributed through out the world for redundancy and performance objectives for their global user base. In terms of building scale, each data center can scale beyond a hundred thousand square meters with 100s of MW of power. Even with the ever growing needs for hyper-scale data centers today, there is a lack of publicly available literature which discuss coupling building energy models with IT operations to help operators reason about actual service workloads and their building's performance

In the recent past, the power usage effectiveness metric (*PUE*) as defined by Equation 1 has been a key performance indicator for DC facilities architects and facilities operators.

$$PUE = \frac{E_{total}}{E_{IT}} \quad (1)$$

$$E_{total} = \text{Total Power Used at Facility}$$

$$E_{IT} = \text{Power Consumed by IT Equipment}$$

The *PUE* is a intuitive metric to reason about. In consideration of the *PUE*, the mechanical inefficiencies and electrical distribution losses external to the IT equipment contribute to the difference between the

numerator and denominator of Equation 1. In operations, the facilities and IT parameters change simultaneously together in a continuous sequence, lending to accurate real time evaluations. However, the continuous states of IT workloads and ambient environmental conditions makes design time PUE calculations complicated. The complications drive DC building designers to evaluate the efficiency metric, by averaging discrete step-wise (hour-day-week) workload values and taking the worst case values to benchmark the facility.

However, without accounting for continuous workload profiles these coarse models misrepresent the part-load operations of the DC and inaccurately quantify total energy demands over a time period. The misrepresentation has significant impacts throughout the life of the hardware deployed in these DC facilities. For example without realistic coincident alignment of workloads and ambient conditions, the thermo-electric equipment whose energy use is dependent on workloads and ambient temperatures may be over-sized for significant portions of their operational times. Yet, during these times the equipment are still allocated their nameplate power draws regardless of their lower operational points. The allocation translates to power being reserved regardless of real demand.

The simulations demonstrated in this research allow for more realistic evaluations of DC energy use. These more realistic evaluations are useful for two things. First, operators can use the realistic cyber-physical models to simulate the DC building performance under perceived usage changes. Second, by exposing the facility’s power and cooling equipment headroom for non-design day conditions they can enable DC operators to opportunistically oversubscribe the IT loads. Such a oversubscribing scheme can be implemented by extending the electrical over-subscription leading to free capacity schemes presented by Li [Li et al., 2018] when coupled with predicative modeling techniques.

The contribution of this work is the demonstration of an IT load aware building energy model that integrates an external Python script to dynamically reset the IT load that must be cooled within EnergyPlus. The rest of the paper discusses specific modules that one of the authors built to couple the building systems and IT.

In the next section, similar works on data center energy models are discussed. In the subsequent section, Methodology, the methods and reasoning behind the modules developed in this work are discussed. The results from implementing these methods are then presented in the Results section followed by a Discussion section in which the authors reason about the article’s results and claims. The paper then concludes by highlighting the contribution and identifying opportunities for extending this work.

Similar Works

As described above, PUE is not a full proof metric for DC energy efficiency. By acknowledging that PUE does not account for IT equipment efficiency, Perolini provides a coordinated control strategy across IT workload and building infrastructure based on network queuing theory [Parolini et al., 2012]. Perolini logically composes a set of IT systems and supporting infrastructure as a singular cyber-physical system network graph. Hierarchically structured quality of service (QoS) values is common among Perolini and Li, who uses the QoS to optimize IT performance in terms of the power distribution system [Li et al., 2018]. Both authors dynamically tune IT with the facilities infrastructure systems. In terms of exploiting the QoS, the fundamental difference between these two works is that Li considers the IT workload as the controlled variable bound by the facility’s power distribution systems and Perolini considers the cooling equipment as the controlled variable bound by the IT workload. In their recognition of lower QoS DC workloads as being preemption tolerant and higher QoS as time sensitive workloads; these methods allow over-subscription of power and thermal capacities without any service level performance degradation.

DC cooling systems have been algorithmically coupled with DC IT workloads by Wei in [Wei et al., 2017]. They use a reinforcement learning agent that is aware of the DC’s thermal state in real-time. The agent acts as a controller of the HVAC equipment and respective IT traffic balancer, where the balancer migrates work between two or more DCs. In the work they demonstrated that when given a state-action pair and an objective function; the agent takes a control action in a given state to reduce power while allowing the IT system to meet it’s performance objective in terms of processing latency. The state-action pairs are optimized with an objective function that values the power demand along with a IT performance metric at a given time-step. Wei’s work was limited to DCs located in mixed use buildings, where the cooling system had some fungibility with other parts of the building, allowing waste energy from the DC to be reclaimed to heat the other parts. Nonetheless, their work is similar to this research in that it also integrated EnergyPlus together with Python.

Methodology

In this section a dynamic workload-profile based energy simulator is introduced and the simulation is described in detail. First, an overview of this work’s cross platform software integration is presented. Then details about each of the two main components, namely EnergyPlus and Python based traffic models, are discussed. As shown in Figure 1, the proposed simulator uses Python to reset the DC IT load for each time-step. In the subsequent modeling time-step, the EnergyPlus models run with the

externally revised variables. The basic process loop of the simulator is illustrated in Figure 2.

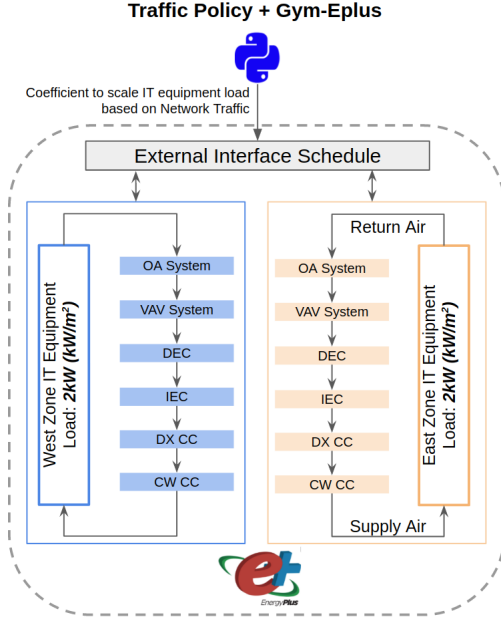


Figure 1: EnergyPlus and Traffic Model.

The simulator integrates algorithms that are aware of network traffic profiles programmed in Python with EnergyPlus through the Building Controls Virtual Test Bed (BCVTB) [Wetter and Noudui, 2016]. The BCVTB interface from Gym-EPlus [Zhang et al., 2019] is used to implement the interaction. This work extends Gym-Eplus capability by making it reset the IT load in DCs in addition to its thermal reset features. DCs have been explicitly supported in EnergyPlus since Version 8.3.0. Using the already supported DC variable for CPU load schedules in EnergyPlus, the xml file configuration as shown in Figure 3 couples the Python module with EnergyPlus.

The baseline EnergyPlus model of the data center is adapted from Moriyama [Moriyama et al., 2018]. Moriyama's building shell consists of two data center rooms. The first room, the West Zone is 232 square meters and the second room, East Zone, is 259 square meters. Each data center room has dedicated mechanical and electrical equipment that are symmetrical to each other. Throughout the simulations the building envelope, mechanical, and electrical modeling objects are not modified from the baseline.

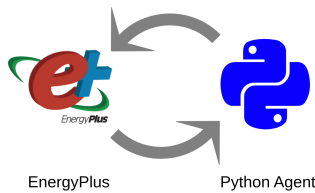


Figure 2: Dynamic Process Flow between EnergyPlus and the Python Agent.

```

1 <?xml version="1.0" encoding="ISO-8859-1"?>
2 <!DOCTYPE BCVTB-variables SYSTEM "variables.dtd">
3
4 <BCVTB-variables>
5
6 <!-- Recieve from E+ -->
7 <variable source="EnergyPlus">
8   <EnergyPlus name="East Zone" type="Zone"
9     Thermostat Heating Setpoint Temperature"/>
10 </variable>
11
12 <variable source="EnergyPlus">
13   <EnergyPlus name="East Zone" type="Zone"
14     Thermostat Cooling Setpoint Temperature"/>
15 </variable>
16
17 <variable source="EnergyPlus">
18   <EnergyPlus name="Whole Building" type="Facility"
19     Total Building Electric Demand Power"/>
20 </variable>
21
22 <variable source="EnergyPlus">
23   <EnergyPlus name="Whole Building" type="Facility"
24     Total Electric Demand Power"/>
25 </variable>
26
27 <!-- Send to E+ -->
28 <variable source="Ptolemy">
29   <EnergyPlus schedule="HeatingSetPoint_RL"/>
30 </variable>
31
32 <variable source="Ptolemy">
33   <EnergyPlus schedule="CoolingSetPoint_RL"/>
34 </variable>
35
36 <variable source="Ptolemy">
37   <EnergyPlus schedule="Data Center CPU Loading
38     Schedule_RL"/>
39 </variable>
40
41 </BCVTB-variables>

```

Figure 3: EnergyPlus variables.cfg used to integrate Python and EnergyPlus. Other variable maps from GymEplus are listed in the variable.cfg, but not used in these simulations.

However, for this research, as the goal is to represent hyper-scale DCs; the IT equipment objects are changed to closer resemble the current practices in this class of DCs based on one of the authors industrial experience.

Three statically configured changes are made to the baseline model's ITE object. Namely the changes are; server inlet temperature, fan power ratio compared relative to the overall server power, and the CPU power density fields are changed. The typical EnergyPlus ITE object for each zone is provided in Figure 4. First on line 16, the server entering air temperature is changed to 25°C from 15°C . Second, on line 12 the fan power fraction is changed to 0.10 from 0.40. The last static variable change increases the load density of the data center to $2 \frac{\text{kW}}{\text{m}^2}$ from $0.2 \frac{\text{kW}}{\text{m}^2}$ on line 8. Details on how this power density was derived is provided later in Algorithm 1.

The only dynamic variable change to the ITE object definition specifies the dynamic CPU Loading Schedule_RL on line 10 of Figure 4. This new schedule replaces the original schedule, CPU Loading Schedule. Both of these schedules are shown in the IDF snippet in Figure 5. The CPU Loading Schedule_RL is called from the IDF as an external interface through the BCVTB. For the BCVTB and EnergyPlus interface, this new schedule is mapped between the two

```

1 ElectricEquipment:ITE:AirCooled,
2 WestDataCenter_Equip,      !- Name
3 West Zone,                  !- Zone Name
4 FlowControlWithApproachTemperatures, !- FC *
5 Watts/Area,                 !- P Calc. Method
6 ,                            !- {W}
7 ,                            !- Number of Units
8 2000,                        !- {W/m2}
9 Data Center Operation Schedule, !- Power Sch.
10 Data Center CPU Loading Schedule_RL, !- cpuSch
11
12 Data Center Servers Power fLoadTemp, !-
13 0.1,                            !- Fan Power Input
14 0.0001,                         !- AFR {m3/s-W}
15 Data Center Servers Airflow fLoadTemp, !-
16 ECM FanPower fFlow,            !- fan Flow
17 25,                             !- EAT {C}
18 A3,                             !- ASHRAE Class
19 AdjustedSupply,                !- Inlet Connection
20 ,                              !- Room Air Model
21 ,                              !- Room Air Model
22 West Zone Inlet Node,          !- SA
23 0.1,                            !- Recirc. Ratio
24 Data Center Recirculation fLoadTemp, !-
25 0.97,                          !- PSU efficiency
26 UPS Efficiency fPLR,           !- UPS PLR Curve
27 1,                              !- PS Losses to Zone
28 ITE-CPU,                       !- CPU End-Use
29 ITE-Fans,                      !- Fan End-Use
30 ITE-UPS,                       !- PS End-Use
31 ,                              !- SAT *
32 -2,                            !- RAT *
33 ,                              !- RAT Schedule *

```

Figure 4: EnergyPlus ITE Object Defintion. New features of EnergyPlus 8.9 are denoted with *.

software platforms in the variable.cfg file on line 38. Then at each time-step, a value for the CPU load is passed from Python to EnergyPlus. The passed value is multiplied by the CPU power density from line 8 of Figure 4 for the simulation time-step.

```

1 Schedule:Compact,
2 Data Center CPU Loading Schedule, !- Name
3 Any Number,                      !- Schedule Type
4 Limits Name                      !-
5 Through: 12/31,                  !- Field 1
6 For: SummerDesignDay,            !- Field 2
7 Until: 24:00,1.00,               !- Field 3
8 For: AllOtherDays,               !- Field 5
9 Until: 6:00,0.50,                !- Field 6
10 Until: 8:00,0.75,               !- Field 8
11 Until: 18:00,1.00,              !- Field 10
12 Until: 24:00,0.80;              !- Field 12
13
14 ExternalInterface:Schedule,
15 Data Center CPU Loading Schedule_RL, !- Name
16 Any Number, !- ScheduleType
17 0; !- Initial value, used during warm-up

```

Figure 5: EnergyPlus IT load reset schedule definitions. The Schedule: Compact is replaced with the schedule defined in the External Interface: Schedule.

There are two ways to input the IT power load into EnergyPlus. The first is the unit count and unit-power method. The second is the power-density method. As seen above the floor area density method is used for this work. The density method is more conducive to scaling with the building size than the unit-power method. Algorithm 1 indicates the steps used to translate the properties from a actual hyper-scale data center to Moriyama’s model. The density (ρ) is simply the product of the total provisions power for the IT equipment on the server floor (P_f) and the total server floor area (A_f).

Although expressing the IT density as a constant value over the entire floor area suffices for EnergyPlus’ view of the IT load as a lumped thermal parameter, it is a naive representation of the power dis-

tribution on the floor. In realty racks are not placed evenly across the entire server floor, but are placed in rectilinear arrangements, as represented by A_r in Algorithm 1 and illustrated in 6. By following the equations in Algorithm ??, the rack counts and rack unit densities can be obtained and the method of entry can be changed.

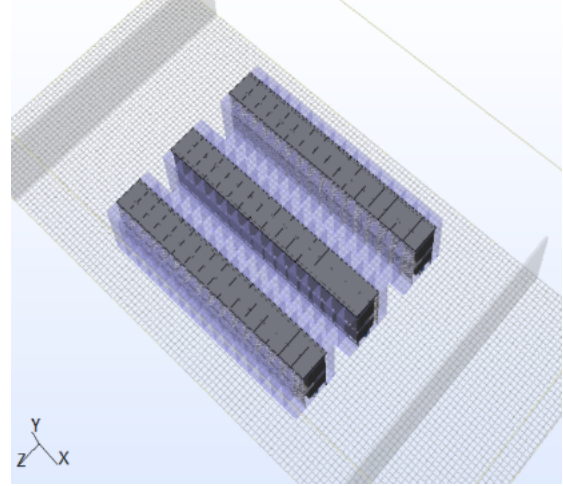


Figure 6: Server room layout. The gray elements in the figure are the hot aisles and the light purple elements are the server racks. The objective of the cited work was to determine the optimal spacing of adjacent racks across hot-aisles. Image from Sahini and Kumar [2016]

Now, the CPU Loading Schedule_RL as passed in from the Python program to EnergyPlus is described. The basis for the schedule is a network traffic model built upon simulations of real world network traffic for a global service. The choice to characterise the DC IT equipment load with bandwidth of network traffic is superior to the usual hour-day-week schedules as an operational model because (1) most DCs serve international users across many time zones. (2) the DC workloads are known to not be stable across time. (3) network traffic is conducive to predicative models based around world events and seasons.

Characterizing the traffic profile for an online service platform requires granular data that spans across a wide temporal range. However, due to the proprietary nature of the internet industry such data that allows for building level traffic profiles is not publicly available. Therefore, in this research a network traffic simulator is created. The objective for the simulator was to represent a globally distributed online platform indicative of real world user demand. To represent a global service, the load factors in this article reflect visits to 145,063 Wikipedia pages over an 18-month period as provided by Kaggle [Kaggle, 2017]. In this work, publicly available information about Wikipedia’s infrastructure establishes the geographical locations for the DCs as illustrated in the map in Figure 7.

Algorithm 1 Rack Counts from Building and Power Properties

$$\begin{aligned}\rho &= \frac{P_f}{A_f} \\ A_r &= A_f \times F_f \\ R_{\#} &= \frac{A_r}{R_a} \\ P_{rs} &= \frac{P_f \times (1 - F_n)}{R_{\#}} \\ P_{rn} &= P_{rs} \times F_n \\ P_r &= P_{rs} + P_{rn}\end{aligned}$$

Where:

ρ = Power Density of server floor. This value is used in the EnergyPlus Models.

A_f = Total floor area.

A_r = Area available for rack deployment.

R_a = Rack footprint area.

$R_{\#}$ = Rack count.

P_f = Provisioned power for IT for server floor.

P_{rs} = Power for rack server equipment.

P_{rn} = Power for rack network equipment.

P_r = Total power of rack.

F_f = Fraction of A_f that is deployable with Racks.

F_n = Fraction of power to rack allocated to network devices.



Figure 7: Data Center Locations Map. SFO: San Francisco, CAR: Carrollton, TX, ASH: Ashburn VA, AMS: Haarlem, Netherlands, SIN: Singapore.

One issue with the Kaggle data set is that it just provides a time series of page visits agnostic of source location (ie user location) or serving DC. For evaluating DC level workloads, the pages were mapped to a DC location as part of a pre-processing step of this research. The pre-processing required handling raw URL strings and parsing out it's language hash markers. Then, countries where these languages are the official languages are mapped to the nearest Wikipedia DC to indicate the traffic to each DC. The step-wise procedure is in indicated in Table 1.

The procedure resulted in the seven languages being routed by a minimum distance function from any country that they are the official language to the nearest DC site. Treating each of the seven languages as a

Table 1: Steps for converting URL to traffic

Step 1:	Map language to page by parsing URL structure.
Step 2:	Map languages to countries where they are the first tongue.
Step 3:	Map ingress sites to countries with minimum distance function.
Step 4:	Get the distribution of the number of countries served by each ingress site.
Step 5:	For each ingress site; sum up the daily views of pages in all language.

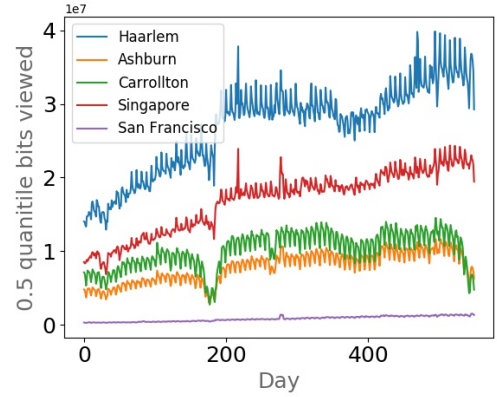


Figure 8: Sum of English, French, German, Mandarin, Spanish, Russian, and Japanese traffic to each of the 5 DC locations.

service provides an abstraction that makes each language an isolated platform.

Figure 8 shows curves with all languages to a particular data center aggregated together. These curves are translated to a coefficient that scales the IT load indicated on line 8 of Figure 4. The coefficients for each data center are normalized to the maximum traffic volume; i.e. max traffic indicates the upper bound with a coefficient of 1. These coefficients are independent across the five data centers.

Next we discuss the results from a model build with the methods discussed in this section.

Results

The traffic aware EnergyPlus models are simulated for each data center location through GymEplus in a batch process loop. In the batch workflow, the same IDF file is used to represent all of the data centers shown on the map in Figure 7; except their local weather files are changed by the scripted loop. In these reset load (RL) simulations, at each time step the traffic load factor is pulled in from the Python program and is passed to EnergyPlus by BCTVB. This demonstrates that using an external interface (such as logic scripted in Python), building energy models can facilitate the analysis of a network of data centers in a single workflow.

A second set of simulations without the external in-

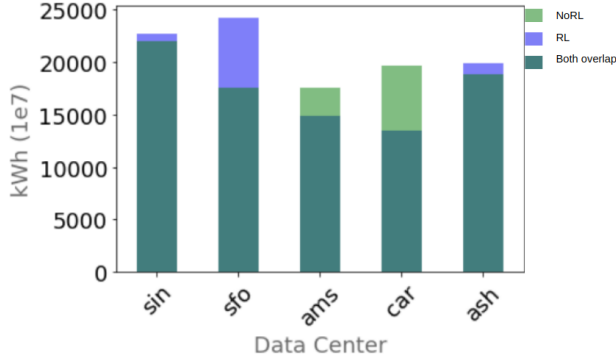


Figure 9: Total Site Energy Difference between statically configured IT load and dynamically reset IT loads.

terface (NoRL) were also run for each DC with a similar workflow to quantify the energy use with default CPU loading schedule shown in Figure 5. The total annualized energy use of the two models is plotted for comparison in Figure 9. The purple bars indicate the simulations in which the CPU loads are reset at each time-step (RL), and the green bars indicate simulations without interactive resetting of the loads (NoRL). For Singapore (sin), San Francisco (sfo), and Ashburn (ash) the RL simulations result in higher annualized energy compared to the NoRL model. While for Amsterdam (ams) and Carrolton, TX (car), the RL model results in lower energy demand compared to the statically configured load from the NoRL simulation.

The variations in the relative energy demands between the two models can be attributed to the CPU load profiles as illustrated in Figure 10. The figure illustrates the time series plot of each DC analyzed through the RL simulations with line curves. The x-axis indicates the hour of the year and the y-axis indicates the CPU load for each respective DC. The CPU profiles for each DC that their average values are distinct from one another. Carrolton and Ashburn appear to have noticeable seasonal patterns. Amsterdam and Singapore DCs show irregular oscillations in demand throughout the year, but have a tight band. The San Francisco DC has a constant line. This constant demand is due to the load not varying much from the peak capacity in the traffic profile that was used for the site.

In contrast to the distinct CPU load profiles of the RL loads as discussed above, with default schedule configuration of IT load profiles used in the NoRL models, the CPU loads at all the DCs are identical to each other. The static property of the NoRL model is also indicated in Figure 10 by the gray band. The gray band is actually composed of line curves that cycle diurnally with the given schedule.

Figure 11 shows a stacked histogram for the set of DCs modeled. This plot shows that even with a

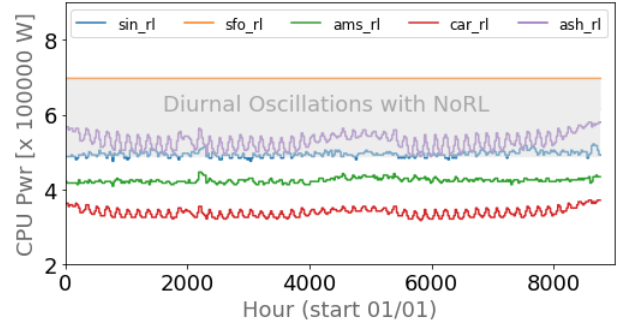


Figure 10: Time series of the total CPU power at each DC (see legend). Gray area indicates the daily demand cycle that the IT equipment creates with NoRL.

identical building systems the efficiency of each DC will vary across the globe. One major contributor to this difference is obviously the local weather conditions which DC operators can't control. However, the second contributor is the workloads at the DC. The workload is a variable that can be controlled by the DC operators.

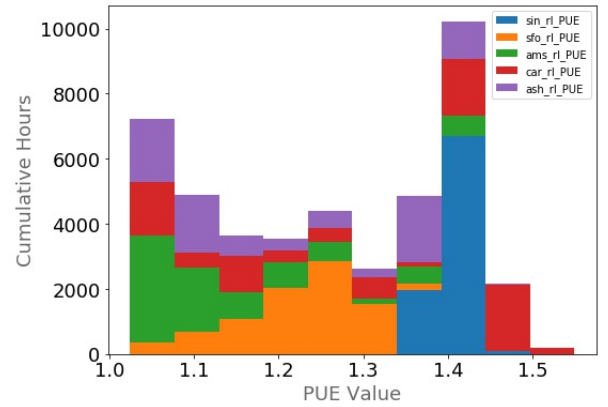


Figure 11: Stacked PUE Histogram for all five data centers .

In the next section, the extensibility of the traffic based building energy model towards predicative scenarios and the potential to use this framework to exploit 'free capacity' are discussed.

Discussion

One obstacle for building operators for using building energy models to make operational decisions is that working with these models requires deep knowledge about the software implementation of the models. The one variable that operators control are the IT loads and allowing the IT loads to be characterized by external software in building energy models can allow operators to do some interesting things. For example, operations teams can simulate perceived future changes to their work loads. As mentioned above predicative modeling is feasible when using network traffic as the proxy for IT loads. Network traffic has been shown to be predictable within reasonable accu-

racy for internet services by LinkedIn zhuang15. Figure 12 shows the predicative models for four of the data center studied in the simulations for this work.

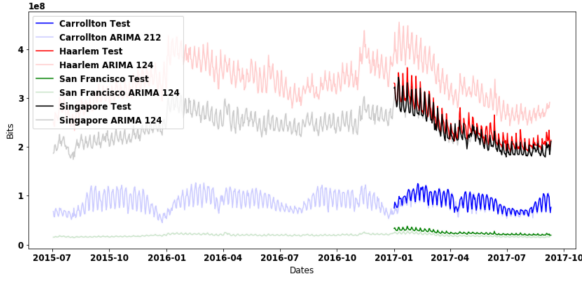


Figure 12: Forecasting Model of the Traffic.

In this research the network traffic serves as a proxy for CPU utilization. In practice the network traffic and CPU have complex correlations that can't be generalized with the presented approach. In production DC environments representative correlations can be obtained between network traffic, the IT workloads the the traffic instantiate, and the workload's CPU demand. However, proving such correlations is out of scope of research. Nonetheless, every data center will be subject to its own traffic models due to it being an intrinsic property of the service.

This research assumes that all the facilities are of the same size in terms of power, space, and cooling. From looking at Figure 8, San Francisco site has a much lower traffic profile than Haarlem. The claim that the same power capacity handles this range of traffic requires another assumption. This assumption is that the CPU-Network ratio vary from site to site. For example a bit of network traffic generates 40 times as much CPU workload in San Francisco as it does in Haarlem. This assumption is the basis for normalizing the traffic coefficient, as described by the discussions on Figure 8. This is a realistic condition in DC fleets were have different generations of IT hardware in operations at various DCs. Or this condition can be encountered when software optimizations to specific workloads are done for targeted consumer markets that are geographically segregated. Nonetheless, since the network coefficient is bound between 0 and 1, and network-workload-CPU model can be supported in the same framework.

As an example of free capacity mentioned above, consider a design day dry-bulb temperature of 105°F. When a peak in IT workload occurs on a more favorable day, say 95°F, the computer room air conditioner heat rejection equipment needs 50% of the designed power value [Liebert, 2020]. With a slight modification in the electrical infrastructure, this difference can opportunistically be allocated to IT workloads. However, slight the the electrical modification, the decision is most cost effectively made during the design phase of the the facility. This influence on the design presents a case for using the fine grained load aware

model presented incpu.load.reset the work during the design phase as-well.

Opportunistically using 50% of the provisioned power from the cooling equipment to power the IT equipment requires some further insights about DC operation. First, the 50% claim is based on nameplate full load amps (FLA) for two different dry-coolers (Models D-466 D-880) are matched with a 105kW indoor computer room air conditioner (DC Model 105). One of the dry-coolers is rated at 95°F and the other is rated at 105°F [Liebert, 2020]. The FLA for these units are 14 Amps and 28 Amps respectively. Physically this translates to breaker sizes, wires sizes, utility power feed reservations, and back-up generator reservations decisions being made using this value. Second, with the claim it is assumed that the power draw has the same proportional scaling with loads cross both sets of units.

To summarize the last two paragraphs more explicitly, in conditions where the ambient enthalpy is lower then the design day, electrical provisions beyond the real-time demand relative to the design values are stranded. This is an important consideration for a building class where its utility is measured by the power delivered to IT equipment use. Anytime when the stranded power can be re-purposed from auxiliary infrastructure (such as chillers, cooling towers, and transformers) to the IT equipment, it leads to compelling return on investment cases.

Conclusion

There are two notable benefits of resetting the IT load values outside of the IDF file. First, defining load values for each hour in the IDF file is toilsome and error prone. The toil of entering values for each simulation time-step would require 8,760 entries in the file. This increases the chance of introducing errors into the simulation. Second, the external definition allows more sophisticated logic to control the time-step variables. In this article, only the IT load factors were reset, none-the-less the same interface can reset many other simulation run-parameters with similar logic. An example of a feedback logic is reinforcement learning, which can be framed to globally optimize the systems and is an topic for future research. The traffic profile used in this article are representative of a globally distributed service's user facing workloads and suffices for purposes of demonstrating the dynamics of DC operational loads. Each real-world service will have a unique profile resulting from a combination of critical and opportunistic back-end workloads. The modular construction of this model is conducive for incorporating the other workloads and being inclusive of back-end workloads.

Figure 8 aggregates all the languages to provide an overall building simulation. However future work that need to assess a service platform can create multiple ITE objects in EnergyPlus and quantify the energy

demands for each in isolation.

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